



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0711-Computer Networks

Cloud Networking Through the OSI Architecture

Presented by:

Thanigaivel S(192565006)

Dept: Cybersecurity

Guided by:

Dr. Shankar Ramalingam

Introduction to Cloud Networking & OSI Model

What is Cloud Networking?

- Cloud networking uses internet-based infrastructure to connect devices, applications, and users.
- It enables scalable, secure, and flexible communication.

What is the OSI Model?

- A **7-layer networking framework** that explains how data travels between devices.
- Helps in designing, troubleshooting, and securing cloud networks.

Why Combine Cloud Networking with OSI?

- Simplifies network management.
- Improves communication between cloud services.
- Enhances security and performance.

OSI Layers in Cloud Networking

OSI Layer	Role in Cloud Networking
Application (7)	Cloud applications (Gmail, Office 365, AWS Console)
Presentation (6)	Data encryption, SSL/TLS, compression
Session (5)	Maintains communication sessions
Transport (4)	Reliable data transfer using TCP/UDP
Network (3)	IP addressing and routing between cloud resources
Data Link (2)	Communication within the same network
Physical (1)	Cables, Wi-Fi, fiber optics, cloud data center hardware

Data Flow in Cloud Networking

Cloud Data Transmission Process

- User opens a cloud application.
- Data is encrypted and formatted.
- A secure session is established.
- TCP/UDP transports the data.
- Routers determine the best path.
- Frames are transmitted through the network.
- Physical devices send data to the cloud server.

Benefits

- Secure communication
- Faster data transfer
- Reliable connectivity
- Easy scalability

Advantages & Conclusion

Advantages

- Scalable cloud infrastructure
- Improved security through encryption
- High availability and reliability
- Efficient troubleshooting using OSI layers
- Better resource utilization

Conclusion

- The **OSI model** provides a structured approach to understanding cloud networking.
- Each layer performs a specific function to ensure reliable communication.
- Cloud providers use these networking principles to deliver secure, efficient, and scalable services.